WORKED SOLUTION

STEP III 2018, Question 3

Pure • Euler Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q3

The key idea

The left-hand side is an Euler (equidimensional) equation: each derivative comes with a matching power of x. Expanding the nested form $(*)$ and comparing coefficients pins down a, b and c. Once the equation is in that form it can be solved by two straightforward integrations, peeling off one derivative at a time; the cases arise exactly when one of those integrals is of x^{-1} and produces a logarithm.

Solution

Throughout we work on $x > 0$, so that all the powers of x are defined.

Finding a, b and c

By the product rule,

$$(x^c y)' = x^c y' + c x^{c-1} y, \qquad \text{so} \qquad x^b (x^c y)' = x^{b+c} y' + c x^{b+c-1} y.$$

Differentiating again,

$$\left(x^b (x^c y)'\right)' = x^{b+c} y'' + (b+c) x^{b+c-1} y' + c x^{b+c-1} y' + c(b+c-1) x^{b+c-2} y.$$

Multiplying by x^a and choosing $a + b + c = 2$ gives

$$x^a \left(x^b (x^c y)'\right)' = x^2 y'' + (b + 2c)\, x y' + c(b + c - 1)\, y.$$

This matches the given left-hand side provided

$$b + 2c = 1 - 2p, \qquad c(b + c - 1) = p^2 - q^2.$$

The first gives $b + c - 1 = -2p - c$, so the second becomes $-c^2 - 2pc = p^2 - q^2$, that is $(c + p)^2 = q^2$. Taking $c + p = q$,

$$c = q - p, \qquad b = 1 - 2p - 2c = 1 - 2q, \qquad a = 2 - b - c = 1 + p + q.$$

So the equation can be written as

$$x^{1+p+q}\left(x^{1-2q}(x^{q-p}y)'\right)' = \mathrm{f}(x),$$

as required. (The other root $c + p = -q$ gives the same thing with q replaced by $-q$, which is no different, since only q^2 appears in the equation.)

Part (i)

With $\mathrm{f} = 0$ we can divide by $x^{1+p+q} \neq 0$:

$$\left(x^{1-2q}(x^{q-p}y)'\right)' = 0 \quad\Longrightarrow\quad x^{1-2q}(x^{q-p}y)' = A \quad\Longrightarrow\quad (x^{q-p}y)' = Ax^{2q-1},$$

for a constant A. The next integration depends on whether the power $2q - 1$ is -1.

? **Ask yourself** Why does the value of p not create separate cases here?

The exponent we integrate is $2q - 1$, which does not involve p; p only enters through the factor x^{q-p}, which we divide by at the end. So the cases depend on q alone.

Case $q \neq 0$. Integrating, $x^{q-p}y = \dfrac{A}{2q}x^{2q} + B$, and multiplying by x^{p-q} (and renaming the constant $\frac{A}{2q}$ as A),

$$y = Ax^{p+q} + Bx^{p-q}.$$

Case $q = 0$. Now $(x^{-p}y)' = Ax^{-1}$, so $x^{-p}y = A\ln x + B$ and

$$y = x^p\,(A\ln x + B).$$

In each case there are two arbitrary constants, as expected for a second-order equation.

Part (ii)

This is the case $q = 0$ with $\mathrm{f}(x) = x^n$, so $a = 1 + p$, $b = 1$, $c = -p$, and the equation reads

$$x^{1+p}\left(x(x^{-p}y)'\right)' = x^n \quad\Longleftrightarrow\quad \left(x(x^{-p}y)'\right)' = x^{n-p-1}.$$

The first integration involves x^{n-p-1}, which is x^{-1} exactly when $n = p$.

Case $n \neq p$. Integrating once,

$$x(x^{-p}y)' = \frac{x^{n-p}}{n-p} + A \quad\Longrightarrow\quad (x^{-p}y)' = \frac{x^{n-p-1}}{n-p} + \frac{A}{x}.$$

Since $n - p \neq 0$ again, integrating gives

$$x^{-p}y = \frac{x^{n-p}}{(n-p)^2} + A\ln x + B, \qquad \text{so} \qquad y = \frac{x^n}{(n-p)^2} + x^p\,(A\ln x + B).$$

Case $n = p$. Now the first integration gives a logarithm:

$$x(x^{-p}y)' = \ln x + A \quad \Longrightarrow \quad (x^{-p}y)' = \frac{\ln x}{x} + \frac{A}{x}.$$

Since $\frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{2}(\ln x)^2 = \frac{\ln x}{x}$,

$$x^{-p}y = \tfrac{1}{2}(\ln x)^2 + A\ln x + B, \qquad \text{so} \qquad y = x^p\left(\tfrac{1}{2}(\ln x)^2 + A\ln x + B\right).$$

In both cases the part with A and B is the complementary function from part (i) with $q = 0$, as it should be.

Checks

- Take $p = q = 0$: the equation is $x^2y'' + xy' = 0$, i.e. $(xy')' = 0$, so $y = A\ln x + B$, matching part (i).
- Try $y = x^m$ in the homogeneous equation: $m(m-1) + (1-2p)m + p^2 - q^2 = (m-p)^2 - q^2$, so $m = p \pm q$, a repeated root when $q = 0$, just as found.
- In part (ii) with $n = p = 1$, $y = \frac{1}{2}x(\ln x)^2$ has $y' = \frac{1}{2}(\ln x)^2 + \ln x$ and $y'' = \frac{\ln x}{x} + \frac{1}{x}$, so $x^2y'' - xy' + y = x\ln x + x - \frac{1}{2}x(\ln x)^2 - x\ln x + \frac{1}{2}x(\ln x)^2 = x$, as required.
- All the solutions were also substituted back into the equation by computer at many random values of p, q, n and x; the residuals were zero to rounding error (around 10^{-6}).

Common slips

- Forgetting the case $q = 0$ in (i), or the case $n = p$ in (ii), where $\int x^{-1}\,\mathrm{d}x = \ln x$ replaces the power rule.
- Losing the constant of integration at the first step: it is what produces the second independent solution.
- Assuming $a + b + c = 2$ without saying why: it is needed so that y'' is multiplied by x^2.

Beyond STEP

Writing a second-order operator as a composition of first-order ones is called *factorising* the operator, and it is the idea behind reduction of order and, in quantum mechanics, ladder operators. The substitution $x = \mathrm{e}^t$ turns any Euler equation into one with constant coefficients, which explains the repeated root $m = p$ and the $\ln x$ (that is, t) factor. Boyce and DiPrima, *Elementary Differential Equations*, treats Euler equations and the method of reduction of order.

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